



Profiling Microbial Diversity and Methanogenesis Genes in Biogas Bioreactors Through Metagenomic and Dynamic Analysis



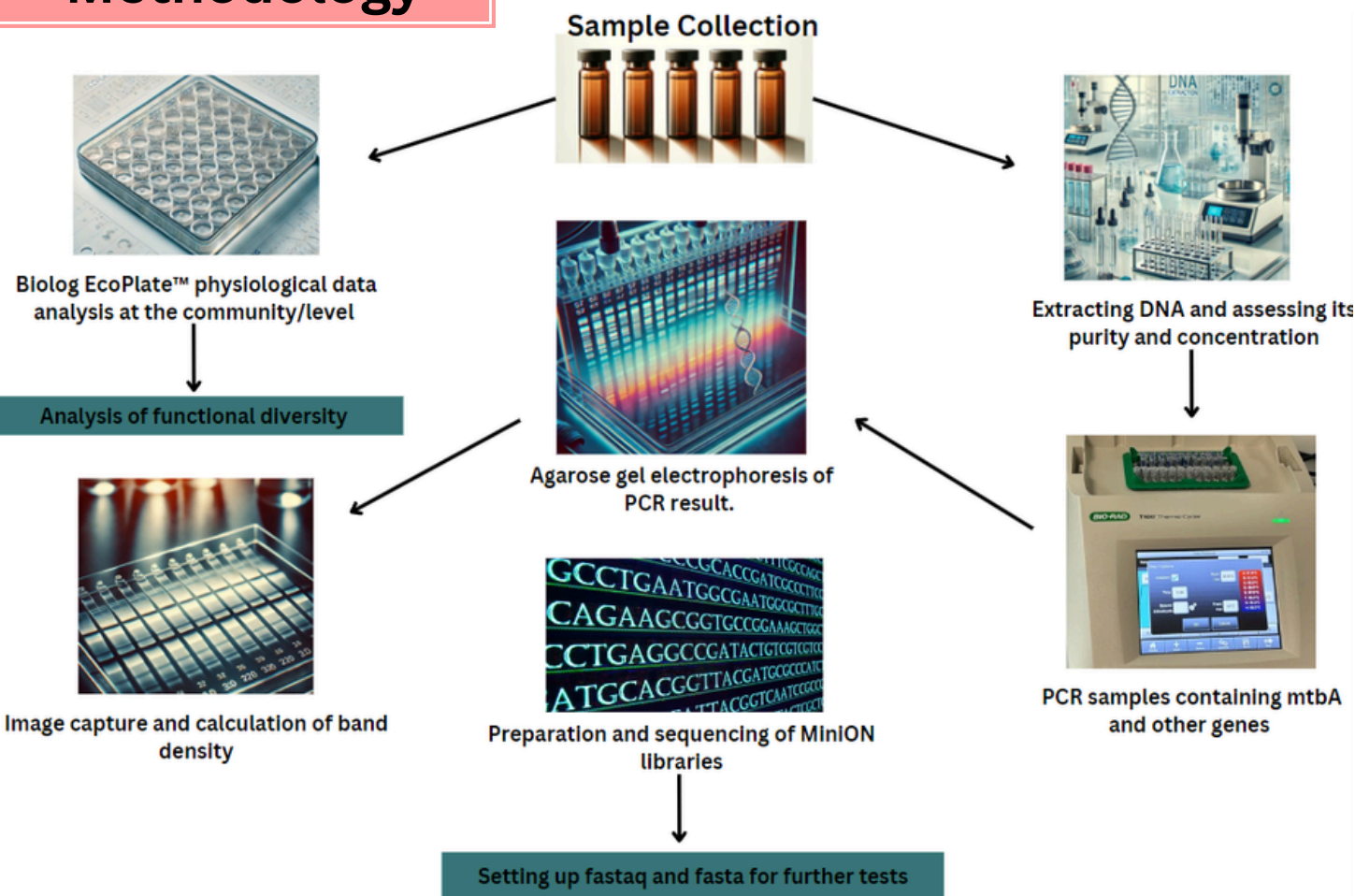
Introduction

Anaerobic digestion, which relies on the activity of complex microbial communities, is the primary driving force behind biogas production. These populations of bacteria and archaea work together to transform organic waste, such as whey, into methane, a renewable energy source. Companies such as Marches Biogas Ltd. have made advances in this industry by deploying microbial consortiums that aid in the breakdown of organic materials into smaller molecules. The methanogenesis pathway, which is essential for methane generation, includes molecules such as methanol, methylamine, and dimethylamine. Microbes process these chemicals and transform them into methyl-coenzyme M (Methyl-CoM), a crucial intermediary in the methane generation chain.

Aim and objective

This study sought to evaluate the changes in microbial communities at various stages of biogas production, with a focus on metabolic activity and diversity. Genes were found using metagenomic analysis to obtain insight into the genetic roles of microbial consortia in biogas reactors, with the goal of increasing methane production efficiency.

Methodology



Analysis of functional diversity

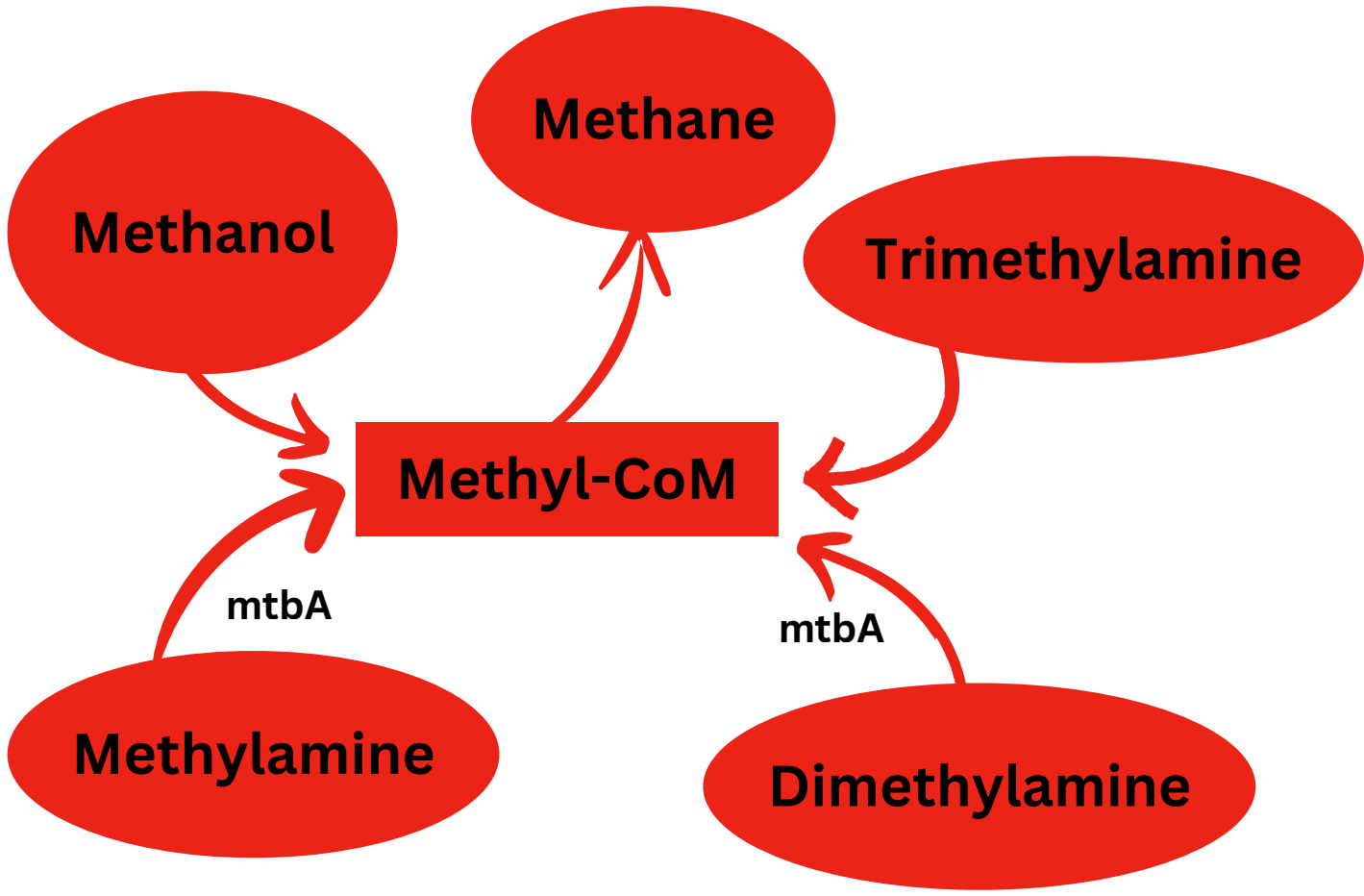
The eco plate is commonly used to compare microbial communities and functional diversity in biogas samples. Most samples exhibit a high functional diversity, indicating robust and resilient microbial eco systems capable of utilizing a variety of carbon sources. Certain samples demonstrated increased activity in methyl-co enzyme M reductase pathways, whereas all samples showed limited efficiency in metabolizing long-chain fatty acids.

PCR samples containing mtbA and other genes

PCR samples containing the mtbA gene and other related methane metabolism genes were used to profile the microbial communities involved in biogas production. The amplification revealed strong bands for mtbA, indicating active methane conversion. Some samples showed high co-expression of mtbA, indicating robust methanogenic activity. Others showed weaker amplification of these genes, indicating lower microbial efficiency in methane metabolism.

References

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Discussion and Result

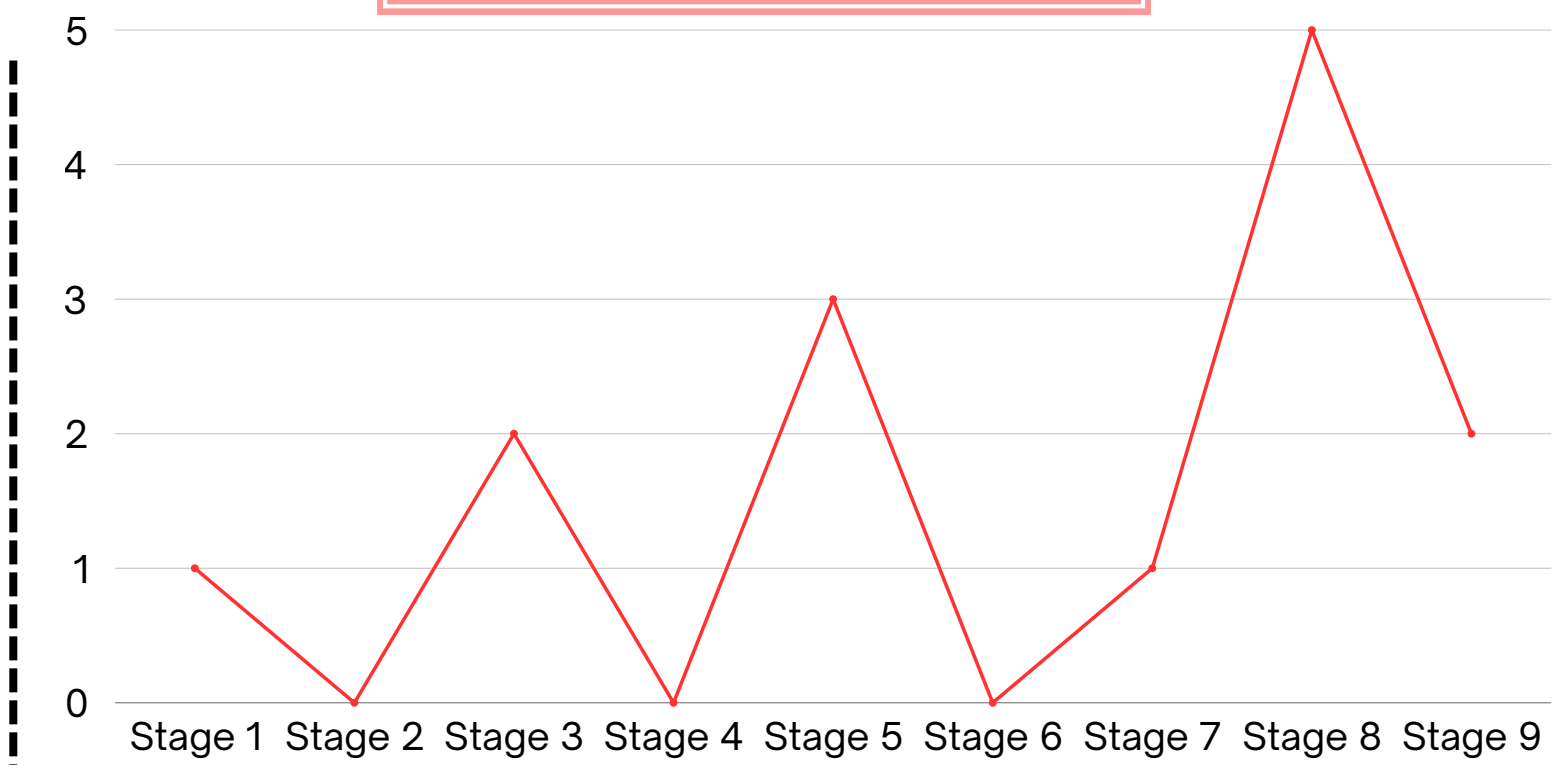


Figure 1: *Methanosarcina barkeri* is identified, indicating the presence of methanogen in the methanogenesis pathway in the sample; apart from *Methanosarcina mazei* and *Methanosarcina acetivorans*, *Methanosarcina barkeri* has a significant dominance.

Gel Analysis

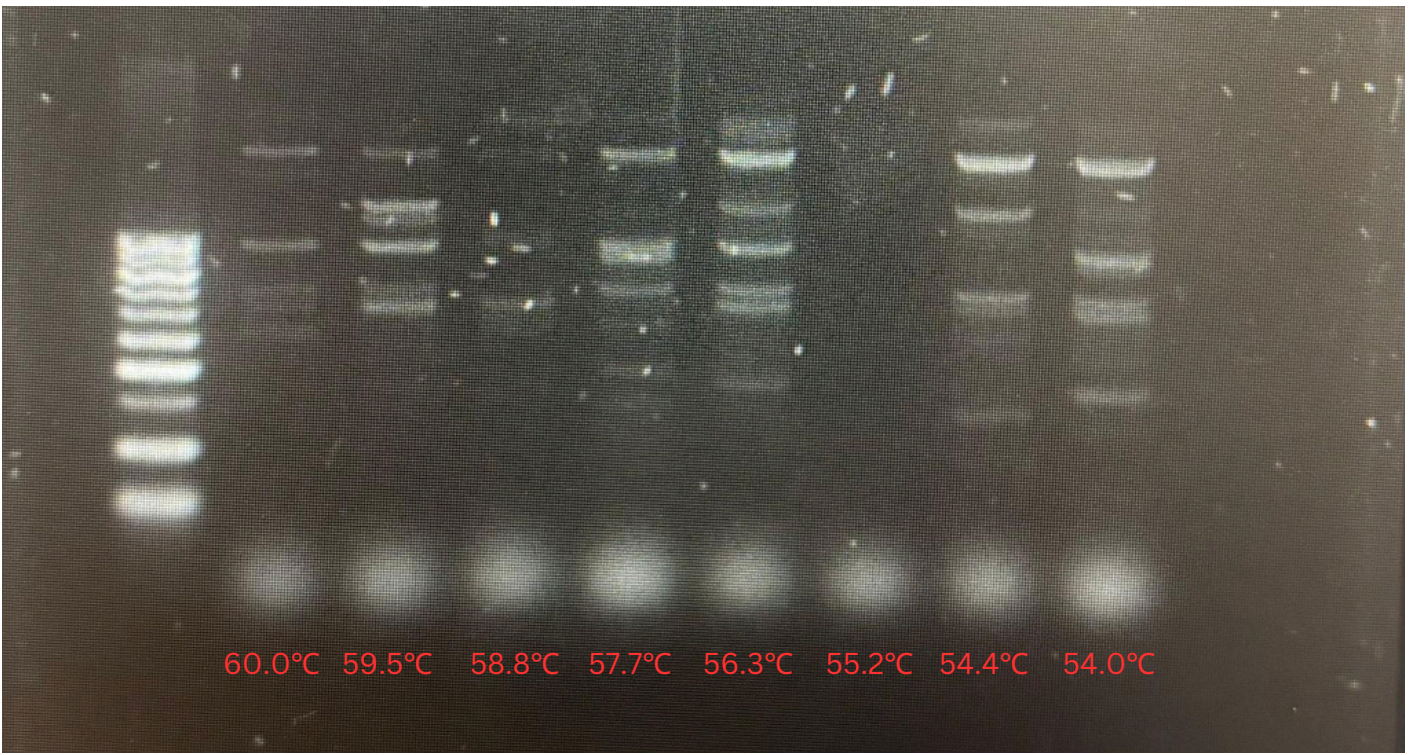


Figure 2: The figure shows that the mtbA gene amplification was unsuccessful, which could be due to a variety of factors. One reason could be suboptimal PCR conditions, such as incorrect annealing temperatures or insufficient primer specificity, which result in inefficient binding. Another possibility is that the extracted DNA has degraded or is of poor quality, which may interfere with the amplification process. The low abundance of the mtbA gene in the microbial community or inhibitors in the DNA samples may have interfered with the PCR reaction, resulting in reduced amplification efficiency.